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Inverse Problems and Imaging (02624)

Week 8

Beyond ordinary Tikhonov regularization

Tikhonov $\|Kx - y\|_{L^2}^2 + \alpha\|x\|_{L^2}^2 \sim \|Ax - y\|_2^2 + \alpha\|x\|_2^2$

Translation $\|Kx - y\|_{L^2}^2 + \alpha\|x - x_0\|_{L^2}^2 \sim \|Ax - y\|_2^2 + \alpha\|x - x_0\|_2^2$

Smoothness Sobolev space $H^1(a, b)$ with norm

$$\|x\|_{H^1(a,b)}^2 = \|x\|_{L^2(a,b)}^2 + \|x'\|_{L^2(a,b)}^2$$

gives "smoothness" regularization

$$\|Kx - y\|_{L^2}^2 + \alpha\|x\|_{H^1}^2 \text{ or just } \|Kx - y\|_{L^2}^2 + \alpha\|x'\|_{L^2}^2$$

The discrete analouge is

$$\|Ax - y\|_2^2 + \alpha\|Lx\|_2^2$$

with L being some Matrix approximation of a derivative (e.g., a finite difference). Could be higher order derivatives.

Total variation functional

For $f \in C^1([a,b])$ (actually just $W^{1,1}(a,b)$)

$$TV(f) = \int_a^b |f'(t)| \, \mathrm{d}t.$$

More generally for $f \in L^1(a,b)$

$$TV(f) = \sup_{v \in C_C^1(a,b),\ |v| \leq 1} \int_a^b f(t) v'(t) \, \mathrm{d}t.$$

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This generalize to higher dimensions $f \in C^1(\overline{\Omega})$ (or $W^{1,1}(\Omega)$)

$$TV(f) = \int_\Omega |\nabla f|\,\mathrm{d}\mu = \sup_{v \in C_C^1(\Omega,\mathbb{R}^n),\ \|v\| \le 1} \int_\Omega f\,\nabla \cdot v\,\mathrm{d}\mu$$

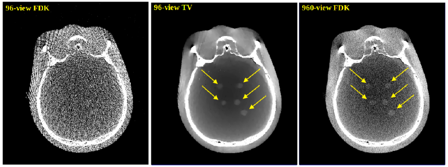
the latter applies to $f \in L^1(\Omega)$.
The space $BV(\Omega) = \{f \in L^1(\Omega) \mid TV(f) < \infty\}$.

TV regularization

$$\|Kx - y\|_{L^2}^2 + \alpha TV(x) \quad \sim \quad \|Ax - y\|_2^2 + \alpha\|L_1 x\|_1 .$$

Original image | Noisy and blurred image | TV deblurred image

From [J. Dahl *et.al.*, Numerical Algorithms, 53 (2010), pp. 67(92)].



[Bian et al. 2010, Phys. Med. Biol. **55**, 6575–6599]. Courtesy: X. Pan, U. Chicago.

Sparsity regularization

With a selected ONB φ_j for X weak for a reconstruction

$$x_{\text{sparse}} = \sum c_j \varphi_j; \qquad c_j = (x_{\text{sparse}}, \varphi_j)_X,$$

with few $c_j \neq 0$.

Under mild conditions such a solution is found by minimizing

$$\|Kx - y\|_{L^2}^2 + \alpha \sum_j |(x, \varphi_j)_X|$$

Electrical Impedance Tomography: Tikhonov vs sparsity

Phantom

Regular Tikhonov

Sparsity

Weighted Sparsity